

## Experimental Psychology 30 - Assignment 4

### Module 2 Section 3 and 4

**/34**

#### Part One: Multiple-Choice Questions (7 marks)

| <i><b>Letter</b></i>                                                                       | <i><b>Statement</b></i>                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>B</b>                                                                                   | <p>1. Infomercials often use _____ to sell products.</p> <p>A. the placebo effect<br/> B. statistical illusions<br/> C. re-interpretation bias<br/> D. regression to the mean</p>                                                                                                                                                                                                  |
| <b>D</b>                                                                                   | <p>2. When the behaviour of people changes simply because they are being observed, the phenomenon is termed the _____ effect.</p> <p>A. bias<br/> <b>B. confounding</b><br/> C. observer<br/> D. Hawthorne</p>                                                                                                                                                                     |
| <b>D</b>                                                                                   | <p>3. Inattentional blindness can cause someone to miss something that is right before their eyes, and this can alter the interpretation of results in psychological research. This was demonstrated in the video about selective attention when many participants did not see the _____.</p> <p>A. umbrella<br/> <b>B. people in black</b><br/> C. chimpanzee<br/> D. gorilla</p> |
| Use the information on Magnets and Pain Relief to answer multiple-choice questions 4 to 7. |                                                                                                                                                                                                                                                                                                                                                                                    |

## ANSWER KEY

|          |                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>B</b> | <p>4. According to the course information, magnets have been used to reduce</p> <ul style="list-style-type: none"><li>A. stomach aches</li><li>B. joint pain</li><li>C. constipation</li><li>D. memory loss</li></ul>                                                                                                                                                                   |
| <b>A</b> | <p>5. Consumers can purchase many items with magnetic properties. According to the course information, which of the following is NOT an item containing magnets?</p> <ul style="list-style-type: none"><li>A. Underwear</li><li>B. Dog collars</li><li>C. Personal massagers</li><li>D. In-soles</li></ul>                                                                              |
| <b>D</b> | <p>6. Dr. Weintraub used 375 participants in his magnetism study. The number of participants was important because</p> <ul style="list-style-type: none"><li>A. it showed how important pain management is</li><li>B. the results would exclude variables</li><li>C. it showed how many people believe in magnetism</li><li>D. the results would be statistically significant</li></ul> |
| <b>A</b> | <p>7. A major problem with Dr. Weintraub's study relates to the fact that participants were able to determine if they received the magnetic insole or the placebo. This means these participants were not</p> <ul style="list-style-type: none"><li>A. blinded</li><li>B. randomized</li><li>C. informed</li><li>D. effected</li></ul>                                                  |

## Part Two: Written-Response Questions (12 marks)

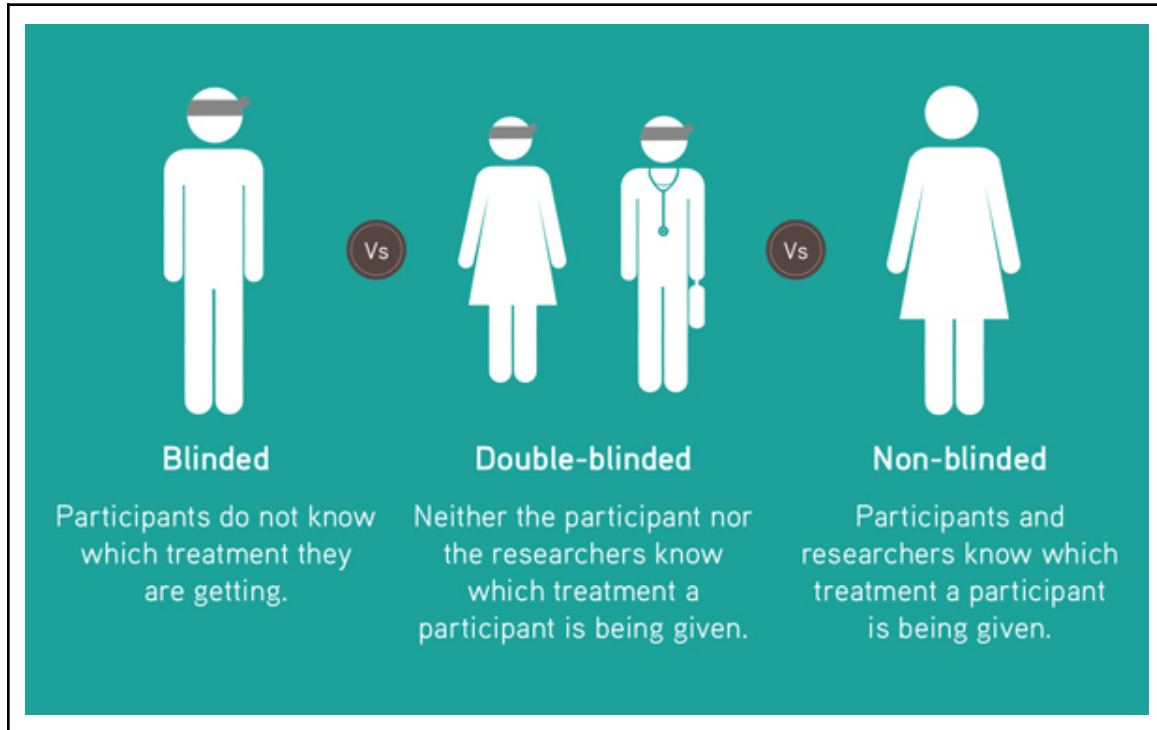
1. What is the importance of randomization when assigning participants to the experimental and control groups? **(2 marks)**

As much as we would like to, we cannot trust the direct personal experiences of people. Confounding factors are partly responsible for this lack of trust because confounding factors can create the illusion that ineffective treatment is actually

## ANSWER KEY

effective. It ensures that neither the researchers nor the participants know who is given the real treatment and who is given the fake treatment or placebo.

2. What is the difference between a single-blind experiment and a double-blind experiment? (2 marks)



3. List the 8 confounding factors discussed in the course information and provide a brief description of how each confounding factor can affect research results (8 marks)

| Confounding Factor           | Effect on Research Results                                                                                                                                                                  |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The Placebo Effect           | The placebo effect, in essence, is a measurable, observable, or perceived improvement in health (physical or psychological) that is not attributable to the treatment.                      |
| The Re-interpretation Effect | more of a perceptual illusion than a real benefit. For example, even when a treatment does not improve symptoms, people may re-interpret their symptoms and experience them as less severe. |

## ANSWER KEY

|                         |                                                                                                                                                                                                                                                                                                                                               |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Observer Bias           | Also known as confirmation bias, observer bias really refers to "selective noticing" whereby researchers tend to notice and to look for what confirms their beliefs and to ignore or overlook the relevance of what contradicts their belief systems.                                                                                         |
| Selection Bias          | If researchers were allowed to choose who receives the legitimate treatment and who receives the placebo, a significant risk would occur that the researchers would subconsciously choose people in such a manner that the outcome of the experiment would be more favourable.                                                                |
| Statistical illusions   | relates to observational studies in which correlation is often confused with causation.                                                                                                                                                                                                                                                       |
| The Hawthorne Effect    | regardless of what condition is changed (or if it was changed at all) people will alter their behaviours simply because they are being observed. People singled out for a study of any kind may improve their performance or behaviour, not because of any specific condition being tested, but simply because of the attention they receive. |
| Experimenter's Bias     | there is a risk that participants will be influenced by the interaction with researchers, not just with the manipulated variable. The outcome of an experiment tends to be biased towards the result expected or desired by the researchers.                                                                                                  |
| Inattentional Blindness | People tend to focus their attention on things they find interesting or important. It is easy, too, to focus too narrowly or on only the negative aspect of events. When our focus is skewed in this way, we do not get the entire "picture" and our complete understanding of events and situations is compromised.                          |

## Part Three: Psychological Research Proposals (15 marks)

## ANSWER KEY

1. Consider the following question: *Does listening to music while studying help students to perform better in exams?*

Would this problem be suitable for conducting psychological research? What criteria did you use to make your decision? **(3 marks)**

Yes, this question would be suitable for psychological research because it focuses on human behaviour, specifically the effect of listening to music while studying on exam performance.

2. Provide 4 reasons why it is important for researchers to prepare research proposals? **(4 marks)**

- to outline the research questions
- to summarize the selected research topic
- to present ways that would be best in conducting the study
- to justify doing the study

3. Using the Topic of Interest Question Checklist, determine if using music to improve performance on exams would be a suitable topic for you. **(8 marks)**

\*remember to provide a statement that indicates whether the topic is suitable for you. You must show evidence of working through the checklist to receive full marks.

- Students must demonstrate that they have worked through the Topic of Interest Question Checklist. (7 marks)
- Provide a statement that indicates whether the topic is suitable for them (1 mark)

----- End of Assignment 4 -----

**When you have completed your assignment, upload this document to the Assignment 4 dropbox on Brightspace.**